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Experiment Title : Demonstration of Switch Hardware and Setting up a Basic Local Area Network

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Experiment No. 06: Demonstration of Switch Hardware and Setting up a Basic Local Area Network (LAN)

Aim

To identify the hardware interfaces of a Network Switch and to design, configure, and verify a basic Local Area Network (LAN) using a Star Topology in Cisco Packet Tracer.

Theory

1. What is a Network Switch?

A **network switch** is a device used in a **Local Area Network (LAN)** to connect multiple devices such as computers, printers, servers, and routers so they can communicate with each other efficiently.

Unlike a hub, a switch is **intelligent** because it sends data **only to the specific device** that needs it, instead of broadcasting to all devices.

Example:
In a college lab, a switch connects all PCs so they can share files, access the internet, and communicate internally.

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2. Main Functions of a Network Switch

A network switch performs several important tasks:

a) Connect devices in a LAN

It acts as a central connection point for devices in the network.

b) Forward data intelligently

A switch sends data only to the intended device using the device's MAC address.

c) Reduce network traffic

By sending data only where needed, it reduces unnecessary traffic and improves performance.

d) Improve network performance

It allows multiple devices to communicate simultaneously without major collisions.

e) Provide security control (basic level)

Switches limit data visibility so only the destination device receives the information.

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3. How does a Switch use MAC Address?

Each device connected to a network has a unique **MAC (Media Access Control) address**, which is a hardware identifier assigned to the network interface card (NIC).

When data is sent through a switch:

1. The switch reads the **source MAC address** of incoming data.
2. It stores this MAC address in its internal table.
3. It checks the **destination MAC address**.
4. It forwards the data **only to the correct port** where the destination device is connected.

This process ensures fast and accurate delivery of data.

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4. What is a MAC Address Table (CAM Table)?

A **MAC address table**, also called a **CAM (Content Addressable Memory) table**, is a list maintained by the switch that maps:

MAC Address → Switch Port Number

Example:

MAC Address	Port
00:1A:2B:3C:4D:5E	Port 1
11:22:33:44:55:66	Port 2
AA:BB:CC:DD:EE:FF	Port 3

The switch uses this table to quickly determine where to send data.

If the MAC address is not found in the table, the switch temporarily sends the data to all ports (except the sender). This is called **flooding**.

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5. Plug-and-Play Nature of Basic Network Switches

Basic network switches are **plug-and-play devices**, meaning they require **little or no manual configuration**.

Key characteristics:

- Devices start communicating as soon as they are connected.
- The switch automatically learns MAC addresses.
- No need to assign IP addresses to the switch for basic operation.
- Easy to install and use.
- Ideal for small networks such as homes, labs, or offices.

Example:
If you connect multiple computers to a switch using Ethernet cables, the switch automatically detects the devices and enables communication without needing special setup.

Hardware Components Required

- PC-PT (x3): End devices acting as network nodes.
- Switch (2960-24TT): A 24-port Cisco Catalyst switch used to interconnect the nodes.
- Copper Straight-Through Cables: The transmission media used for PC-to-Switch connectivity.

Hardware and Connection Setup

- Network ID: 192.168.1.0
- Subnet Mask: 255.255.255.0

Note: PC Naming Convention:

- PC 1: Batch_Roll No._A (e.g., S11_01_A)
- PC 2: Batch_Roll No._B (e.g., S11_01_B)
- PC 3: Batch_Roll No._C (e.g., S11_01_C)

Procedure

Step 1: Physical Topology Design

- Open Cisco Packet Tracer and select Network Devices > Switches > 2960. Place it in the center.
- Select End Devices > PC and place three units around the switch.

- Naming: Click on each PC label and rename them following the convention. (e.g. S11_01_A, S11_01_B, and S11_01_C).

Step 2: Hardware Identification

- Click on the Switch and select the Physical tab.
- Zoom in to observe the 24 FastEthernet ports (10/100 Mbps) and 2 GigabitEthernet ports (1000 Mbps).
- Note the status LEDs and the Console port used for administrative access.
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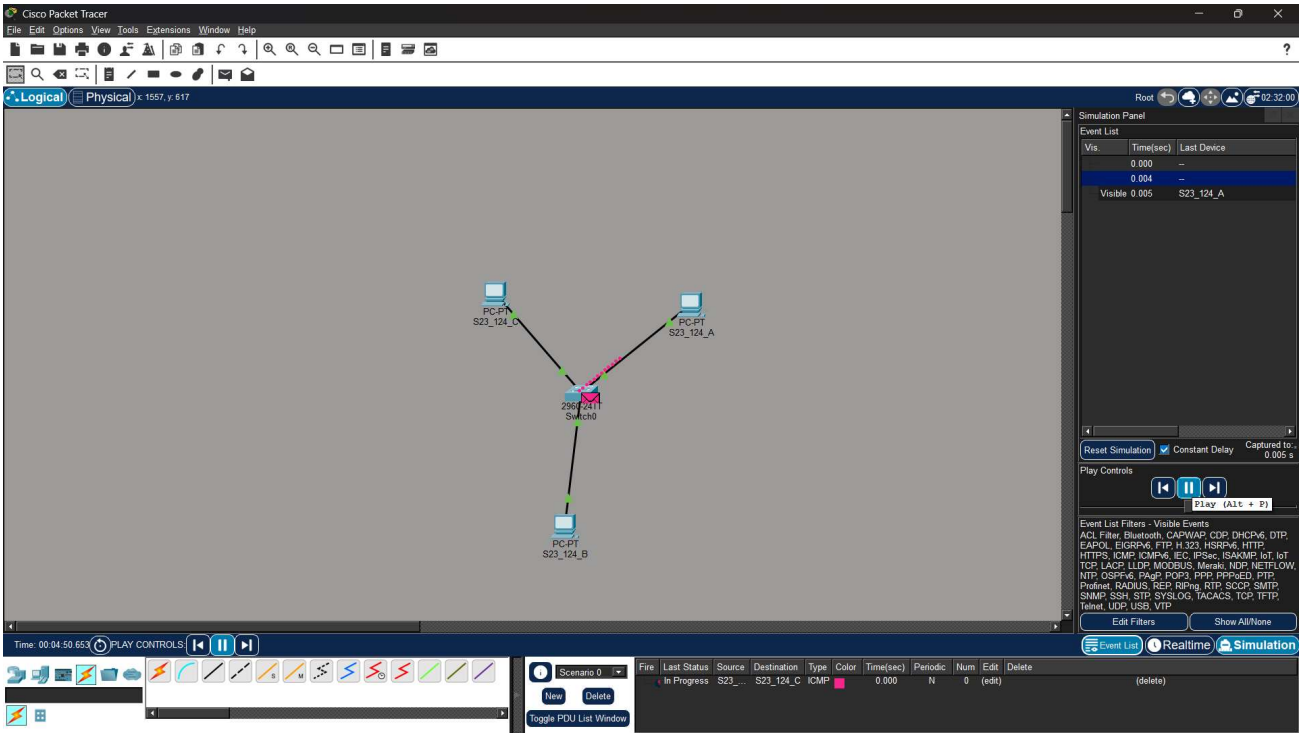
Step 3: Physical Cabling

- Select the Copper Straight-Through cable (solid black line).
- Connect S11_01_A (FastEthernet 0) to Switch (FastEthernet 0/1).
- Repeat for the other two PCs using Switch ports 0/2 and 0/3.
- Observation: Notice the link lights are Orange. Click the "Fast Forward Time" button to turn them Green, indicating the Spanning Tree Protocol (STP) has cleared the ports for traffic.

Step 4: Logical Addressing (IP Configuration)

- Click S11_01_A > Desktop > IP Configuration.
- Enter Static IP: 192.168.1.1 and Subnet: 255.255.255.0.
- Configure S11_01_B with IP 192.168.1.2 and S11_01_C with IP 192.168.1.3.**Step 5: Testing and Simulation**
- **Real-time:**
    - Open the Command Prompt on S11_01_A and type ping 192.168.1.3. A "Reply from..." message confirms the LAN is functional.
- **Simulation Mode:**
    - Click the Simulation tab and click Edit Filters to select only ICMP.
    - Use the Add Simple PDU (Envelope) to click S11_01_A then S11_01_C.
    - Click Play and observe the Switch forwarding the frame only to the destination PC.



Observations

- Media Transition: The Switch ports require time to transition from "Learning" (Orange) to "Forwarding" (Green).
- Micro-segmentation: During simulation, it was observed that the Switch does not broadcast the ICMP packet to all ports; it uses the MAC address table to send it directly to the intended recipient.
- Star Topology Logic: Disconnecting one PC did not affect the communication between the other two, proving the robustness of the Star Topology.

Conclusion

The experiment successfully identifies the hardware features of a Switch and demonstrates the setup of a functional LAN. It concludes that a Switch is the central component of a Star Topology, providing efficient, collision-free data transfer within a single local subnet.